SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I K. YATHEESH (192412669) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks
Concept Identification	20%	2	2	2
Concept Mapping Structure	25%	2	2	2
Linking & Relationships	20%	1	1	1
Application & Analysis	20%	1	1	1
Explanation & Clarity	15%	1	1	1

Total Marks 24 /30

Signature of the Faculty

CONCEPT MAPPING QUESTIONS

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3.Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

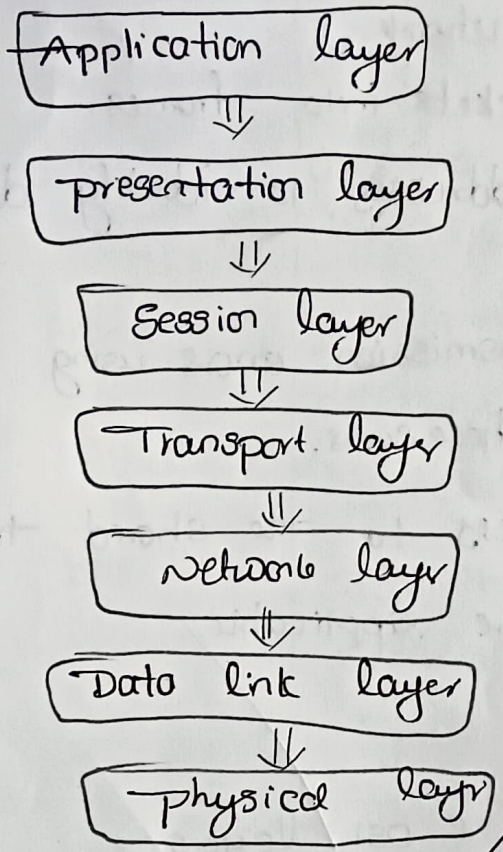
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

CC-1 (Concept) (19) | K. VATHEESH
Assessment Tool - 2 | 192412669
CSAC710 - computer networks

Missing OSI layers & their functions:-

=> The OSI model consists of seven layers that help computers communicate over a network.



Functions of Missing layers:-

=> Session layer:-

=> establishes communication sessions b/w two devices (or) applications.

=> Maintains & manages active sessions during data exchange.

=> provides synchronization points & helps communication after interruptions.

=> Ensures that communication b/w applications on organized process.

Data link layer:-

=> Transfers data b/w directly connected devices on the same network.

=> converts packets into frames.

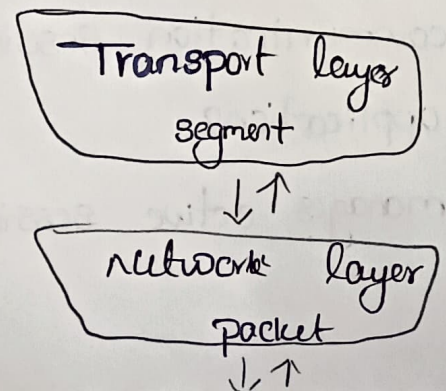
=> uses MAC addresses to identify devices on a local network.

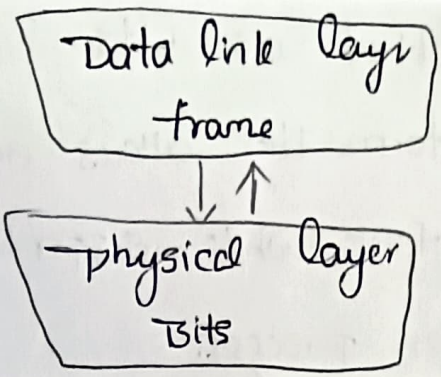
=> Detects transmission errors using mechanism such as frame check sums.

=> Controls access to the shared transmission medium where applicable

Data Units & OSI layers :-

=> Each OSI layer handles data in a specific format called a protocol Data Unit.





OSI layer	Data unit	Function
Transport	Segment	provides end-to-end transport Sequencing & Reliability when using TCP.
Network	packet	uses IP adresses to Route data b/w networks
Data link	Frame	uses MAC addresses for local delivery & supports error detection.
physical	Bits	Transmits binary data as Electrical, optical (or) radio Signals

=> The Transformation of data units is called encapsulation at the Sender & decapsulation at the Receiver

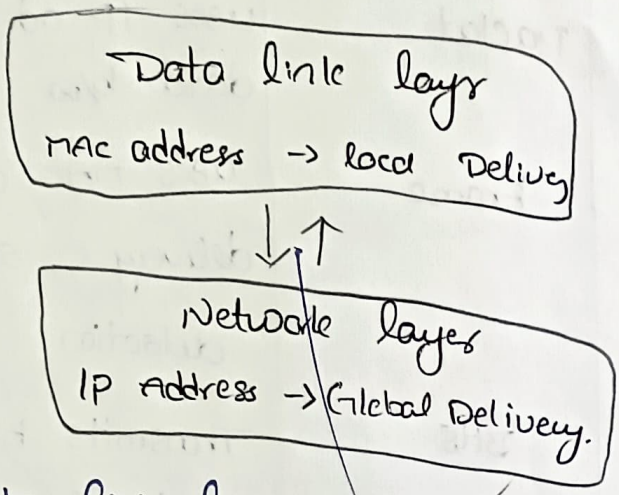
=> when sending a photograph over the internet the data is transported in segments or datagrams carried in IP Packets, placed inside link-layer

frames & transmitted as bits

=> Thus, data transformation allows each layer perform it's own task while supporting the complete transmission process..

=> UDP uses a datagram rather than a segment at the transport layer.

MAC Address - IP Address :-



=> The Data link layer & network layer work together to deliver Data from the source Device to the destination device even when they belong to different networks

=> Data link layer :-

=> The Data link layer operates mainly within a local network such as home college or office network.

MAC address identifies a network interface on the local link..

=> A college computer sends a frame to a nearby router using the Router's local MAC address.

=> Network layer:-

=> The network layer provides logical addressing using IPv4 (or) IPv6 addresses

=> IP addresses identify the source & destination at the network layer

=> A college computer sends a packet to web server located in another country. Routes forward the packet across networks based on it's destination IP address

